

Empirical Proof Record: VALIDATED

2026-05-23T23:14:37.297085+00:00

Empirical Proof Record — VALIDATED

Proof ID: 7a8e068466d606dfed784d544066d76d...

Created: 2026-05-23T23:14:37.297085+00:00

1. Claim

Under sustained load on a balanced text corpus, Kimera's entity target completes each cycle in under 5.00 seconds at the 95th percentile (the architectural throughput ceiling).

- **Operationalisation:** 95th-percentile of per-cycle wall-time (raw['cycle_seconds']) across the streamed batch; cycles that failed to record a cycle_seconds are excluded from the denominator
- **Threshold:** p95_cycle_wall_time_s <= 5.0 seconds
- **H0:** p95 wall-time > 5.00s
- **H1:** p95 wall-time <= 5.00s

2. Verdict

VALIDATED — observed 3.8971680900198398

measured per-cycle wall-time on 40/40 cycles (0 adapter errors); distribution: median 1.49s, p95 3.90s, p99 5.81s (range 0.25-6.86s); batch totals: wall 86.43s, cpu 90.02s (periodic_subprocess_sampler), rss_peak 3820.0MB, max threads 96, max procs 3

3. Evidence

Pillar	Statistic	Value	95th Percentile
0.engineering.p95_wall_time	p95_cycle_wall_time_s	3.8971680900198398	
0.engineering.wall_time_distribution	cycle_wall_time_distribution	1.4903680626302958	
0.engineering.batch_cpu	batch_cpu_total_s	90.01924266399999	
0.engineering.rss_peak	rss_peak_bytes	3820027904.0	

4. Pre-registration

- Registered at: 2026-05-23T23:13:10.629945+00:00
- Config hash: 311bd781c294b3d83c2235bf6767fba8...
- Data hash: 83109a27c3df2a45b912398b969fa8f7...

5. Signature

119ebb4fea67aef36e78003be21eade59d3976ef62c1a019edb750446cf5b604